SDST: Self-Supervised Double-Structure Transformer for Hyperspectral Images Clustering

Fulin Luo, *Senior Member, IEEE*, Yi Liu, Yule Duan, Tan Guo, *Member, IEEE*, Lefei Zhang, *Senior Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Due to the lack of labeled information and the high spectral variability in high-dimensional hyperspectral images (HSIs), HSI clustering has emerged as an effective unsupervised approach for HSI information extraction and classification. Deep clustering methods have achieved significant success in unsupervised HSI classification (HSIC) and have gained increasing attention. However, these methods have limitations in terms of robustness, adaptability, and feature representation when dealing with complex large-scale HSI datasets. Therefore, this article introduced a novel self-supervised double-structure transformer (SDST) approach for HSI clustering. Specifically, in our approach, we designed a shared autoformer structure based on an autoencoder to learn the global properties of HSI data by fusing the multilevel features from the autoencoder with the transformer. Furthermore, we proposed a Siamese dual-former graph module (DFGM) with superpixel-level features for fewer nodes, which reveals long-dependency graph convolutional features, resulting in more precise graph structure features. By constructing a graph with long-dependencies, this module significantly preserves the properties of global dependencies, while focusing on the local features of each superpixel to better represent the fine-grained local details. Finally, we designed a joint optimization module to jointly optimize the double-structure model composed of the shared autoformer module and the Siamese DFGM. To validate the effectiveness of the proposed SDST method, we conducted a series of experiments on the Salinas, Botswana, Indian Pines, and Houston2013 datasets. The proposed SDST achieves competitive clustering accuracies compared with the state-of-the-art clustering methods. The codes are available at https://github.com/YiLiu1999/SDST.**

Manuscript received 11 December 2023; revised 7 February 2024; accepted 29 February 2024. Date of publication 12 March 2024; date of current version 21 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62371076, Grant 62071340, Grant 62201109, and Grant 42201371; in part by the Natural Science Foundation of Chongqing under Grant CSTB2022NSCQ-MSX0452; in part by the Key Research Projects of Universities in Henan Province under Grant 23A510015; and in part by the Fundamental Research Funds for the Central Universities under Grant 2023CDJXY-039. *(Corresponding author: Yule Duan.)*

Fulin Luo and Yi Liu are with the College of Computer Science, Chongqing University, Chongqing 400044, China (e-mail: luoflyn@163.com; 202214131160@stu.cqu.edu.cn).

Yule Duan is with the College of Information Science and Engineering, Henan University of Technology, Zhenzhou 450001, China (e-mail: duanyule@haut.edu.cn).

Tan Guo is with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: guot@cqupt.edu.cn).

Lefei Zhang and Bo Du are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: zhanglefei@whu.edu.cn; dubo@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3374597

***Index Terms*— Autoencoder, graph convolution, hyperspectral image (HSI) clustering, self-supervised learning, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) are captured by a hyperspectral sensor with a nanoscale spectral resolution for the imaging of the Earth's surface [1]. HSIs provide rich spectral information, enabling to detection of the subtle differences between ground objects, having wide applications in the fields of mineral exploration [2], vegetation monitoring [3], and military reconnaissance [4]. However, the high dimensionality and complexity of HSI data will pose challenges for its interpretation and analysis. Traditional interpretation methods often rely on a large number of high-quality labeled samples, which is time-consuming, labor-intensive, and costly, particularly in remote and uninhabited regions. Therefore, unsupervised land-cover classification analysis has become an important research to overcome these limitations. Clustering, as an unsupervised pattern recognition technique, has been widely applied in HSI interpretation. HSI clustering aims to group similar pixels and separate dissimilar ones by fully exploiting the spectral characteristics of HSI data [5].

The classic methods are based on prototypical features for clustering [6], such as k-means [7] and spectral clustering [8]. These methods directly use original data for clustering, generally containing a large amount of redundant information, resulting in poor clustering results. To reduce the redundant interference, the subspace-based clustering methods [9] have been proposed to map the high-dimensional data into a low-dimensional feature space including principal component analysis (PCA), nonnegative matrix factorization (NMF), sparse subspace clustering (SSC), and low-rank representation (LRR). Based on these classic subspace clustering methods [10], many excellent algorithms have also been proposed for clustering, such as generalized PCA (GPCA) [11], incomplete multiview clustering with reconstructed views (IMCRVs) [12], and robust subspace segmentation by LRR [13]. However, these methods ignore the intrinsic relationship of data, so graph-based subspace clustering methods were proposed, such as subspace clustering via sparse graph regularization (SGRSC) [14], multiview structured graph learning (MSGL) [15], graph-Laplacian correlated LRR (GCLRR) [16], and subspace clustering via learning an adaptive low-rank graph (SC-LALRG) [17]. Graph-based subspace clustering

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

has also been widely used in HSIs. Huang et al. [18] proposed a novel dictionary learning-based subspace clustering method (IDLSC), which learns a common intrinsic cluster from view-specific subspace representations by a new decomposition-based scheme. Liu and Wang [19] proposed a graph convolutional optimal transport (GCOT), which uses optimal transport probability to find the edges of the graph and characterize the local nonlinear structure of the original features. Cai et al. [20] presented a graph convolutional subspace clustering (GCSC) framework, developing the traditional subspace clustering into the non-Euclidean domain. Huang et al. [21] proposed a novel algorithm called bipartite graph partition with graph NMF (BGP-GNMF) to capture the intrinsic relationship between hyperspectral pixels and superpixel-based dictionary atoms. Huang et al. [22] proposed a novel graph convolutional sparse subspace coclustering (GCSSC) model with nonnegative orthogonal factorization for large-scale HSIs, in which affinity matrix learning and spectral coclustering were integrated into a unified optimizing model. However, these methods cannot fully learn the intrinsic features of HSIs and are highly sensitive to noise and outliers [23], which may lead to inaccurate clustering results.

In recent years, the advancement of deep learning techniques has consequently presented new opportunities for HSI clustering [24]. Deep clustering methods utilize deep neural networks to extract complex hidden features from HSIs and divide pixels into clusters. Deep clustering methods aim to combine deep representation learning with clustering tasks [25]. Deep clustering network (DCN) [26] is a joint dimensionality reduction and k-means clustering approach, in which dimensionality reduction is accomplished via learning a deep autoencoder. The deep embedded clustering (DEC) [27] method uses a deep neural network to learn deep representation of data, and then clusters the features with k-means, where the entire network is co-optimized by iteratively refining the deep representation as well as the clustering task. Variational autoencoder (VAE) [28] simulates the generative latent representation to approximate the posterior distribution for clustering. Deep clustering has also achieved great advantages in HSI and a large number of DCNs have been proposed in recent years. Zhao and Li [29] designed a novel algorithm, termed deep spectral clustering with regularized linear embedding (DSCRLE), to benefit from both spectral graph theory and deep learning technique. Mei and Yin [30] proposed a cascade residual capsule network (CRCN) to extract deep spatial–spectral features and introduced the coding rate reduction to measure the compactness of learned spatial–spectral features. Shahi et al. [31] proposed a multisensor deep clustering (MDC) for the joint processing of multisource imaging data. Sun et al. [32] adopted intraclass distance as a constraint condition and proposed an intraclass distance-constrained deep clustering algorithm for HSIs (DCIDC). However, these deep clustering methods do not consider the intrinsic relationship of HSI data.

To analyze the intrinsic relationship of data, graph convolutional networks (GCNs) [33] encode graph structure and node properties for achieving node representation. Bo et al. [34] proposed a structural DCN (SDCN), which attempts to integrate the structural information into deep clustering. Zeng et al. [35] designed GraphSCC by using GCNs to represent the structure relationships between samples. In HSI, similar ground objects are distributed in different areas, and there are long-dependencies between samples. However, graph convolution algorithms have difficulty in capturing long dependencies of HSI.

To obtain long-dependencies of data, transformer [36] models (such as ViT [37], Swin transformer [38], and CrossFormer [39]) are widely used in image classification. Transformer also has advantages in clustering. Nguyen et al. [40] proposed a Clusformer, which realizes automatic visual clustering through an unsupervised attention mechanism. Wang et al. [41] proposed a transformer-based clustering, which unifies the transformer and the contrastive learning for video face clustering. Transformer has also been used in HSI clustering. Xue et al. [42] proposed S2FL, which is capable of extracting high-level meaningful features from multiview data. Li et al. [43] proposed a feature self-relation method, which leverages the feature self-relations for training self-supervised ViT. Liu et al. [44] proposed an excellent self-supervised learning of spectral masking for HSI classification (HSIC), which employs masked spectral reconstruction to construct a self-supervised learning model for pretraining an encoder and decoder composed of multilayer transformers. These transformer models use a single structure model to learn features. However, HSI is complex data that combines space and spectrum. It is difficult for a single structure network to fully mine the intrinsic features of HSI.

To solve these problems, we fully consider the feature association of HSI in different layers and utilize the advantages of the transformer to extract the long-dependency features of HSI. We proposed a novel self-supervised double-structure HSI clustering method called Self-supervised Double-Structure Transformer (SDST). SDST mainly contains three modules: autoformer module, dual-former graph module (DFGM), and joint optimization module. First, we designed a shared autoformer based on the autoencoder and transformer, which achieves the multilevel feature co-representation from autoencoder layers with transformer. Subsequently, we innovatively designed a Siamese dual-former graph structure, which uses superpixels as nodes in graph structure to construct a Siamese graph convolution representation with transformer for achieving long-dependency graph association features. Finally, we designed a joint optimization module to jointly optimize the shared autoformer module and the Siamese DFGM.

In summary, we highlight the main contributions of this article as follows.

1) We designed a shared autoformer structure based on autoencoder and transformer, which learns the global properties of HSI data by fusing the multilevel features from autoencoder with transformer.
2) We proposed a Siamese DFGM, which reveals long-dependency graph convolutional features, resulting in more precise graph structure features.
3) A joint optimization module was developed based on self-supervised learning, which jointly optimizes the shared autoformer module and the Siamese DFGM to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

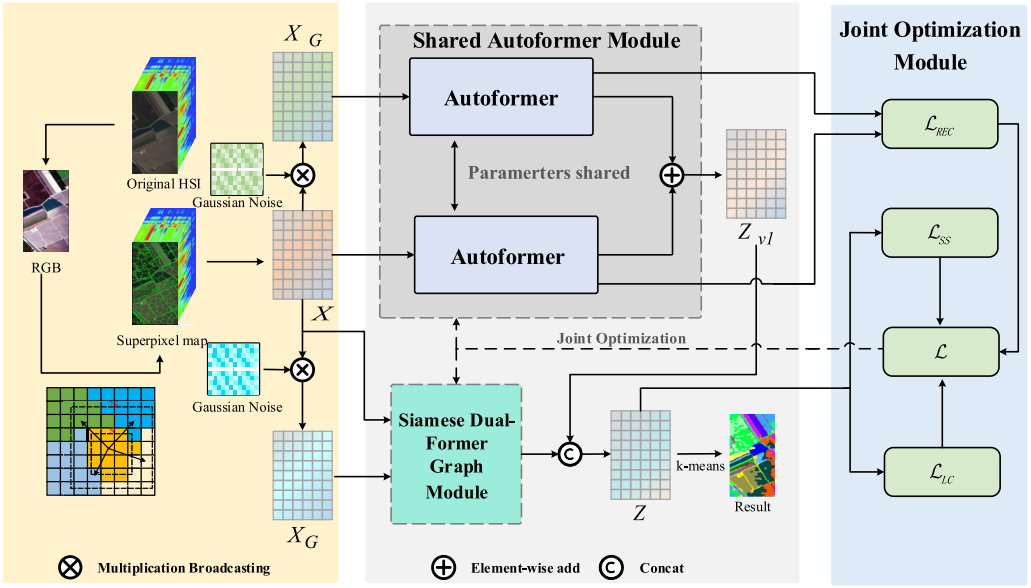


Fig. 1. Architecture of the proposed SDST for HSI clustering. We first perform superpixel segmentation on the RGB channels of the HSI to obtain pixel-level spectral–spatial features. Next, we design a shared autoencoder that is capable of capturing multilevel features from the HSI data. Subsequently, we introduce the Siamese DFGM, which treats superpixels as nodes in a graph. This module reveals long-dependency graph convolutional features. Finally, we incorporate self-supervised learning to cluster the graph nodes and update the weights through joint optimization of the double-structure model during network training.

enhance the representation performance of features for clustering.

The structure of this article is as follows. Section II describes the proposed SDST method in detail. Section III conducts some experiments on four real HSI datasets to demonstrate the effectiveness of SDST. In Section IV, we analyze some conclusion of this article and suggestions for future work.

II. Self-Supervised Double-Structure Transformer

In this section, we introduce the SDST. First, we design a double-structure feature extractor that efficiently integrates both structural and global information [45]. Then, we propose a self-supervised learning method to train the clustering model.

A. Architecture

The overall framework of the proposed SDST is illustrated in Fig. 1. HSI clustering is performed to assign each pixel to different clusters, thereby compacting the intracluster and separating intercluster distance between clusters. In SDST, it contains a shared autoformer module, a Siamese DFGM, and a joint optimization module. The first module uses a shared autoencoder network to learn the multilevel features and transformer to fully fuse the features from different levels. The second module utilizes a Siamese structure based on GCN and transformer to fully extract the correlation information between HSI features. Subsequently, based on self-supervised learning, a joint optimization module is designed to simultaneously train the shared autoformer module and the Siamese DFGM in the SDST framework. For the graph module, directly using HSI pixels as graph nodes will bring a large computational cost. Thus, it needs to reduce the number of graph nodes for the graph model. Superpixels are small regions composed of spatially adjacent pixels that are similar in attributes like color, brightness, and texture. Each region generally contains the same category of land-cover, which does not influence the boundaries of land covers in HSI. Using superpixels instead of a lot of pixels can efficiently decrease the complexity of image analysis. The superpixel segmentation algorithm [46] utilizes spatial information of pixels to group them into blocks with certain internal similarities, reducing image details while preserving essential structural information. Here, we first employ the SLIC superpixel segmentation algorithm [47] on the RGB channels of HSI, which not only preserves the local spatial features but also significantly reduces the number of graph nodes. The RGB channels are used to form visible natural images, thus, for visualization and simplification, we select the wavelength of red (R), green (G), and blue (B) in the HSI spectral for achieving superpixel segmentation. For an HSI $\mathcal{I}$, it can be expressed by superpixels as follows:

$$\mathcal{I} = \cup_{i=1}^{N} S_i,\ S_i \cap S_j = \emptyset, \quad i \neq j; i, j = 1, 2, \ldots, N \tag{1}$$

where $S_i = \{p_{i,1}, \ldots, p_{i,n_i}\}$ represents a superpixel block consisting of n_i interconnected pixels, and N is the total number of superpixels. Compared with ordinary RGB images, HSIs have rich spectral information. However, during the transformation of pixels from RGB images, the essential spectral features are lost [48]. To retain the spectral characteristics of HSI, we directly utilize the spectral features of each pixel from the original HSI and then compute the average spectral value in each superpixel as the superpixel features. Specifically, the spectral features $X_i \in \mathbb{R}^{n_i \times B}$ in a superpixel block from the original HSI $\mathcal{I} \in \mathbb{R}^{H \times W \times B}$, where H, W, and B represent the height, width, and band number, respectively. The superpixel feature X as the input of SDST is calculated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

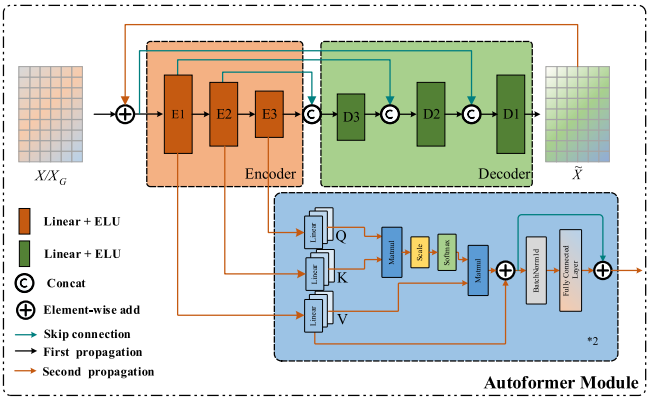


Fig. 2. Structure of autoformer module.

as follows:

$$X = [X_1, X_2, \ldots, X_N]^T = \left[\frac{1}{n_1}\sum_{k=1}^{n_1} x_k^1, \frac{1}{n_2}\sum_{k=1}^{n_2} x_k^2, \ldots, \frac{1}{n_N}\sum_{k=1}^{n_N} x_k^N\right]^T \quad (2)$$

where X_i is the feature vector of the i-th superpixel and x_k^l is the k-th pixel in the l-th superpixel block.

B. Shared Autoformer Module

To extract rich information from HSI, we employ an autoencoder to capture features. Then, we construct a transformer module to achieve the interaction of multilevel features from different layers, which can improve the diverse and discriminative features, as illustrated in Fig. 2. To enhance the robustness of autoencoders, we add random noise into the input features X by a noise matrix $G \in \mathbb{R}^{N\times B}$ with a Gaussian distribution $\mathcal{N}(1, 0.1)$. Subsequently, we obtain the noise features $X_{\mathcal{G}} \in \mathbb{R}^{N\times B}$ with multiplicative noise G, which can be expressed as follows:

$$X_{\mathcal{G}} = X \odot G. \quad (3)$$

Our autoencoder module can be divided into two parts: the feature reconstruction and the multilevel feature fusion.

1) Feature Reconstruction: For a given data feature sample X, the encoder part can transform the spectral information into a fixed-size vector representation. Specifically, the output $H_e^{(l)}$ of the l-th encoding layer can be represented as follows:

$$H_e^{(l)} = \phi\left(W_e^l H_e^{(l-1)} + b_e^l\right) \quad (4)$$

where ϕ denotes the ELU activation function and W_e^l and b_e^l are the weight and bias parameters, respectively. Subsequently, the input features X and the noise features $X_{\mathcal{G}}$ are, respectively, employed as the dual inputs $H_e^{(0)}$ for the encoder of the shared autoformer module. The decoder layers are computed as follows:

$$H_d^{(l)} = \phi\left(W_d^l H_d^{(l-1)} + b_d^l\right) \quad (5)$$

where ϕ denotes the ELU activation function and W_d^l and b_d^l are the weight and bias parameters, respectively.

In deep autoencoders, deeper encoding layers might not always retain all the information from the original inputs. With skip connections, information from shallower layers can be directly transmitted to the decoders, ensuring that crucial details and features are preserved. Therefore, we use a skip connection, directly connecting the features of the encoder and decoder at different layers, to facilitate more effective information transmission and reduce feature loss. Consequently, the inputs of each decoder layer can be represented as follows:

$$H_d^{(l)} = \left[H_e^{(l-1)} \middle|\middle| H_d^{(l-1)}\right]. \quad (6)$$

2) Multilevel Feature Fusion: The encoder transforms the input data into a lower dimensional representation, facilitating the automatic learning of latent features [49]. During the multilevel feature fusion stage, a fully connected layer serves as the initial layer of the network. To be specific, three fully connection operations, i.e., $F_Q(\cdot)$, $F_K(\cdot)$, and $F_V(\cdot)$, are used to obtain the queries Q, the keys K, and the values V, as follows:

$$\begin{aligned} Q &= F_Q\left(H_e^{(l-1)}\right) \\ K &= F_K\left(H_e^{(l)}\right) \\ V &= F_V\left(H_e^{(l+1)}\right) \end{aligned} \quad (7)$$

where $Q \in \mathbb{R}^{n_q\times d_q}$, $K \in \mathbb{R}^{n_k\times d_k}$, $V \in \mathbb{R}^{n_v\times d_v}$. Due to the subsequent calculation of attention, it should ensure $d_q = d_k = d_v$ and $n_k = n_v$. We compute the attention as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q\cdot K^\top}{\sqrt{d_k}}\right)\cdot V. \quad (8)$$

Subsequently, the model employs multihead attention to capture the global properties and the multilevel feature interaction of data. Assuming the attention consists of h heads, the outputs are

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^O \\ \text{head}_i &= \text{Attention}\left(Q W_i^Q, K W_i^K, V W_i^V\right) \end{aligned} \quad (9)$$

where W_i^Q, W_i^K, and W_i^V are the projection weights of the i-th head. W^O is the weight for multihead concatenation.

Finally, the outputs of multihead attention are processed by normalization and full connection layer transformation, with an added residual connection [50], to generate the module's outputs, denoted as $Z_{v_1} \in \mathbb{R}^{N\times S}$.

C. Siamese DFGM

Graphormer [51] represents a successful application of the transformer architecture in graph neural networks [52]. By harnessing the self-attention mechanism in transformers, Graphormer adaptively captures the intricate global dependencies among nodes within a graph. Although Graphormer excels in capturing the overall data information, it might exhibit a relatively diminished ability to capture local details compared with certain traditional GNN models. Particularly, in the tasks that emphasize meticulous local details, the impact of this limitation becomes very significant. Hence, for this influence, we introduce a novel Graphormer module in Fig. 3, termed as Siamese DFGM. This module uses a Siamese structure to process the original HSI and the HSI with Gaussian noise, respectively. Through active noise addition, the Siamese

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

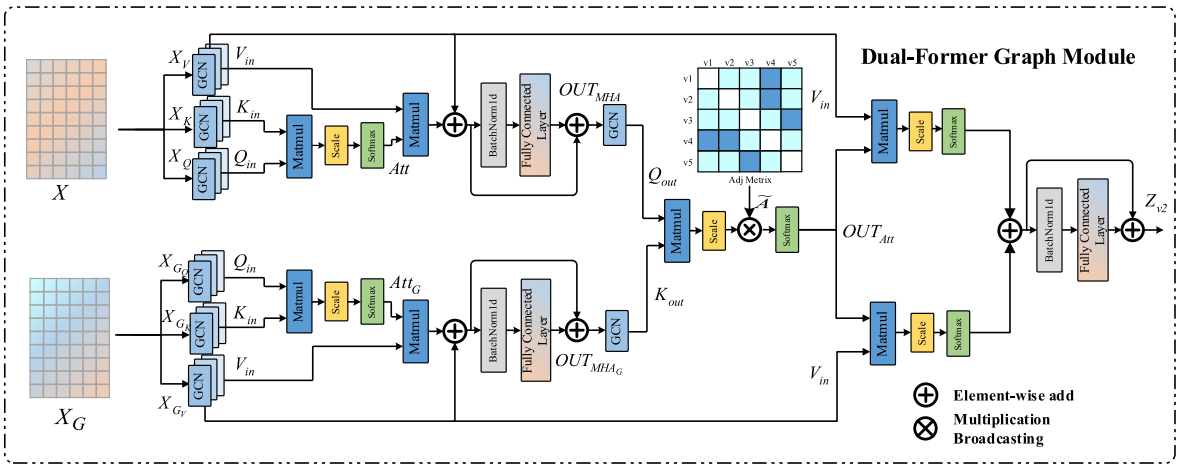


Fig. 3. Structure of DFGM.

DFGM can learn how to identify and ignore noise, thereby improving the network's interference with noise and the robustness of the module. Subsequent experiments also prove that this dual-structure learning improves feature extraction ability. Also in this module, superpixels are employed as the nodes in the graph structure. The internode similarity S is calculated by a radial basis function (RBF)

$$s_{ij} = e^{-\rho \|X_i - X_j\|^2} \tag{10}$$

where X_i and X_j are the embedded features of superpixels i and j, $\rho = 0.2$ is an empirical set to control the RBF width. Based on the similarity matrix $S \in \mathbb{R}^{N\times N}$, we select k most similar superpixels for each individual superpixel as its neighbors to construct the adjacency matrix A

$$A_{ij} = \begin{cases} 1, & \text{if } S_i \in N_k(S_j) \text{ or } S_j \in N_k(S_i) \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $N_k(S_i)$ is the k-nearest neighbors of S_i. The corresponding degree matrix is represented as $D = \text{diag}(d_1, d_2, \ldots, d_N) \in \mathbb{R}^{N\times N}$, $d_i = k$. According to D, the initial adjacency matrix A can be normalized as $\tilde{A} \in \mathbb{R}^{N\times N}$ by computing $D^{-1}(A+I)$, and $I \in \mathbb{R}^{N\times N}$ signifies an identity matrix.

In the Siamese DFGM, these GCNs are employed to obtain queries Q_{in}, keys K_{in}, and values V_{in} as follows:

$$\begin{aligned} Q_{\text{in}} &= \varphi\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}X_Q W_Q\right) \\ K_{\text{in}} &= \varphi\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}X_K W_K\right) \\ V_{\text{in}} &= \varphi\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}X_V W_V\right) \end{aligned} \tag{12}$$

where $Q_{\text{in}} \in \mathbb{R}^{n_q\times d'_q}$, $K_{\text{in}} \in \mathbb{R}^{n_k\times d'_k}$, $V_{\text{in}} \in \mathbb{R}^{n_v\times d'_v}$, and φ denotes the activation function Tanh.

In our DFGM, the obtained Q_{in}, K_{in}, and V_{in}, once acquired, are individually utilized as queries, keys, and values. Then, the scaled dot-product attention mechanism is used to compute the global spatial attention matrix $\text{Att} \in \mathbb{R}^{N\times N}$. For $\text{Att}_G \in \mathbb{R}^{N\times N}$, it represents the attention coefficients from the data augmented with noise. Subsequently, we employ a fully connected feed-forward network to nonlinearly transform and extract features from the attention outputs, OUT_{MHA} and $\text{OUT}_{\text{MHA}_G}$. Furthermore, we apply two additional layers of GCN layers to individually aggregate the structural information for the attention representations, OUT_{MHA} and $\text{OUT}_{\text{MHA}_G}$. These aggregated representations serve as the respective queries Q_{out}, and keys K_{out} for the subsequent modules. Given the interrelatedness and dependency among nodes, we have innovatively integrated the adjacency relationships of the graph into attention. This integration capitalizes on the adjacency matrix $\tilde{A}$, reflecting the interconnections between nodes in the graph, and the attention coefficient matrix, revealing the importance weights among nodes. Their amalgamation facilitates more precise information propagation and feature updates. The model concurrently considers local neighbor information and global associative relationships during the feature update of nodes. This allows for a more effective fusion of information across different areas. The process is formalized as follows:

$$\text{OUT}_{\text{Att}} = \text{softmax}\left(\frac{\tilde{A} \odot Q_{\text{out}} \cdot K_{\text{out}}^T}{\sqrt{d_k}}\right) \tag{13}$$

where $\tilde{A} \in \mathbb{R}^{N\times N}$, $Q_{\text{out}} \in \mathbb{R}^{n_q\times d_q}$, $K_{\text{out}} \in \mathbb{R}^{n_k\times d_k}$. Thereafter, we apply the scaled dot-product attention, sourced from the transformer module, to weigh the initial features of values V_{in}. Ultimately, the weighted values are inputted into a fully connected neural network (accompanied by residual connections) for a linear combination, achieving the final output $Z_{v_2} \in \mathbb{R}^{N\times S}$.

Finally, to obtain richer information, we merge latent embeddings from two different perspectives through a linear combination process facilitated by a fully connected neural network architecture. This orchestrated process yields the clustering-centric latent embedding $Z \in \mathbb{R}^{N\times S}$, which is beneficial to achieving clustering tasks via the k-means algorithm

$$Z = \text{softmax}(W[Z_{v_1}||Z_{v_2}] + b). \tag{14}$$

Technically, the proposed SDST model systematically integrates the attributes and structural characteristics of data, both at the sample level and feature level. This approach effectively retains more discriminative features in the latent space, thereby enabling the network to acquire meaningful representations and subsequently enhancing the performance of clustering.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

D. Joint Optimization Module

To achieve data clustering, we employ the joint optimization module to effectively integrate both the shared autoformer module and the Siamese DFGM for joint training.

1) Self-Supervised Module: In the unsupervised clustering tasks, the foremost challenge lies in the absence of labels to guide network training. To overcome the above problems, we formulate a self-supervised learning optimization module as a strategic remedy. Beyond the conventional reconstruction loss, we introduce the latent representations into the self-supervised learning optimization module. According to parametric t-SNE [53], it utilizes statistical properties of data to identify and accentuate the similarities among data points while simultaneously distancing dissimilar ones, which can better reveal the statistical similarity of data automatically. Specifically, for the i-th sample and j-th cluster, we employ the t-SNE as a kernel to measure the similarity between the data representation z_i and the cluster centroid vector μ_j in the following:

$$q_{ij} = \frac{\left(1 + \left\|z_i - \mu_j\right\|^2/\alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_k \left(1 + \|z_i - \mu_k\|^2/\alpha\right)^{-\frac{\alpha+1}{2}}} \tag{15}$$

where α corresponds to the degrees of freedom in the t-SNE (α is set to 1 in this article), and q_{ij} is regarded as the probability of the i-th superpixel belonging to the j-th cluster, i.e., a soft assignment.

After obtaining the distribution Q for all samples, our objective is to optimize data representation by learning from high-confidence assignments. Specifically, we aim to make data representations closer to cluster centers, thereby enhancing cluster compactness. Therefore, we compute the target distribution P as follows:

$$p_{ij} = \frac{q_{ij}^2/f_j}{\sum_k q_{ik}^2/f_j'} \tag{16}$$

where $f_j = \sum_i q_{ij}$ is the soft cluster frequency. In the target distribution P, each assignment in Q is squared and normalized, aiming to enhance the confidence level of tasks. The weights of SDST are now learned by calculating the Kullback–Leibler divergence between the joint probability target distribution P and Q, which is achieved by

$$\mathcal{L}_{\mathrm{SS}} = \mathrm{KL}(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}. \tag{17}$$

Notably, the target distribution P is computed based on the distribution Q, and P guides the update of Q, thereby assisting the SDST module to improve the feature representation for clustering. This process can be regarded as a self-supervised mechanism, which encourages data representation to align more closely with cluster centroids.

2) Low-Frequency Constraint Module: Given that our approach solely considers the k-nearest neighbors for each sample, the adjacency matrix inherently displays sparsity, with only k nonzero elements in each row/column. However, it is imperative that the final outputs retain local invariance and show representational consistency. First of all, local invariance allows the model to maintain the stability of the outputs under small local changes caused by viewing angle and lighting. So maintaining local invariance can make our network insensitive to these changes, improving the model's robustness and generalization. Second, the representational consistency can ensure the coherence and reliability of our model when inputting different data, and help our model better learn the inherent properties of data, improving the clustering accuracy. Therefore, we focus on minimizing the following objective function, consequently enabling efficient feature extraction:

$$\mathcal{L}_{\mathrm{LC}} = \frac{1}{2N^2} \sum_{ij}^{N} A_{ij} s_{ij} \left\|z_i - z_j\right\|_2^2 \tag{18}$$

where s_{ij} denotes the elements of the affinity matrix S, as referenced in (10).

The designed low-frequency constraint module can be understood as the objective function of normalized Laplacian spectral clustering, which can make the similar nodes i and j in the affinity matrix S close to each other in the new feature space, thereby promoting the feature representation by maintaining the local structure in the new feature space. To set $B_{ij} = A_{ij} s_{ij}$, (18) can be denoted as follows:

$$\begin{aligned}
\mathcal{L}_{\mathrm{LC}} &= \frac{1}{2N^2} \sum_{ij}^{N} B_{ij} \left\|z_i - z_j\right\|_2^2 \\
&= \frac{1}{2N^2} \sum_{ij}^{N} B_{ij} \left(z_i^T z_i - 2 z_i^T z_j + z_j^T z_j\right) \\
&= \frac{1}{N^2} \sum_{ij}^{N} B_{ij} \left(z_i^T z_i - z_i^T z_j\right) \\
&= \frac{1}{N^2} Tr(Z^T (D - B) Z)
\end{aligned} \tag{19}$$

where $D = \mathrm{diag}([\sum_j B_{ij}]_i)$ is a diagonal matrix revealing the degree of vertices, and $B = [B_{ij}]_{i,j}$.

The Laplacian matrix is denoted as $L = D - B$. To normalize the features, we set $Z^T Z = I$, and the normalized Laplacian matrix can be defined as

$$L_{\mathrm{norm}} = D^{-\frac{1}{2}} L D^{-\frac{1}{2}}. \tag{20}$$

Then, the objective function is transformed as follows:

$$\mathcal{L}_{\mathrm{LC}} = \frac{1}{N^2} Tr(Z^T L_{\mathrm{norm}} Z). \tag{21}$$

3) Reconstruction Constraint Module: In our proposed approach, we first employ SDST to reduce HSI data to a low-dimensional latent representation, then apply the k-means algorithm for clustering. However, the mapping from HSI data to the latent space is intricate in practice. Thus, jointly optimizing these tasks could significantly enhance their performance. Here, we introduce the Reconstruction Constraint Module to recover a "cluster-friendly" latent representation and improve data clustering [26]

$$\mathcal{L}_{\mathrm{REC}} = \frac{1}{2N} \sum_{i=1}^{N} \left\|x_i - \tilde{x}_i\right\| = \frac{1}{2N} \left\|X - \tilde{X}\right\| \tag{22}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

where $\mathcal{L}_{\text{REC}}$ represents the reconstruction loss and N indicates the total number of samples.

In summary, the self-supervised module promotes the tight alignment of data representation with cluster centers. The low-frequency constraint module maintains the local invariance and representation consistency of features. The reconstruction constraint module aims to restore latent representations to improve data clustering performance. Based on these losses, we construct a joint optimization module to collaboratively train our SDST, which enhances the feature representation for clustering. Our overall loss function is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{SS}} + \alpha \mathcal{L}_{\text{LC}} + \beta \mathcal{L}_{\text{REC}} \tag{23}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are two hyperparameters to balance the contributions between local structure preservation and clustering.

III. Experimental Results and Analysis

In this section, four benchmark datasets are employed to analyze our proposed network. Subsequently, an in-depth exposition of the experimental results is provided by meticulous analysis.

A. Experiment Setup

The SDST model was implemented in Python (version 3.10.6) using the PyTorch framework. The computational environment was conducted on a computer with an NVIDIA 3090 24G GPU. The SDST framework comprises three distinctive modules: the shared autoformer module, the Siamese DFGM, and the joint optimization module. In particular, the autoformer module integrates an autoencoder along with two layers of transformer. In the feature reconstruction, the autoencoder's dimensions were configured as d-64-256-512, where d is the input dimension. In the multilevel feature fusion, the number of heads for the two transformer layers was set to 3 and 6, respectively, and all the activation functions are exponential linear units (ELUs). In the Siamese DFGM, the number of heads was set as 3. In the joint optimization module, the hyperparameters α and β were set to 1 and 0.1. The model was trained by the Adam optimizer with an initial learning rate of 0.0001.

B. Dataset Description

In our experimental analysis, we employed four well-established HSI datasets: Salinas, Botswana, Indian Pines, and Houston2013. These datasets are used to comprehensively assess the performance of our proposed method.

The first dataset (Salinas) was collected using a 224-band Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the Salinas Valley, CA, USA. With a high spatial resolution (3.7-m pixels), this dataset covers an area of 512 rows and 217 columns. Twenty water absorption bands were excluded from the dataset, including [108–112], [154–167], and 224. The ground truth comprises 16 categories.

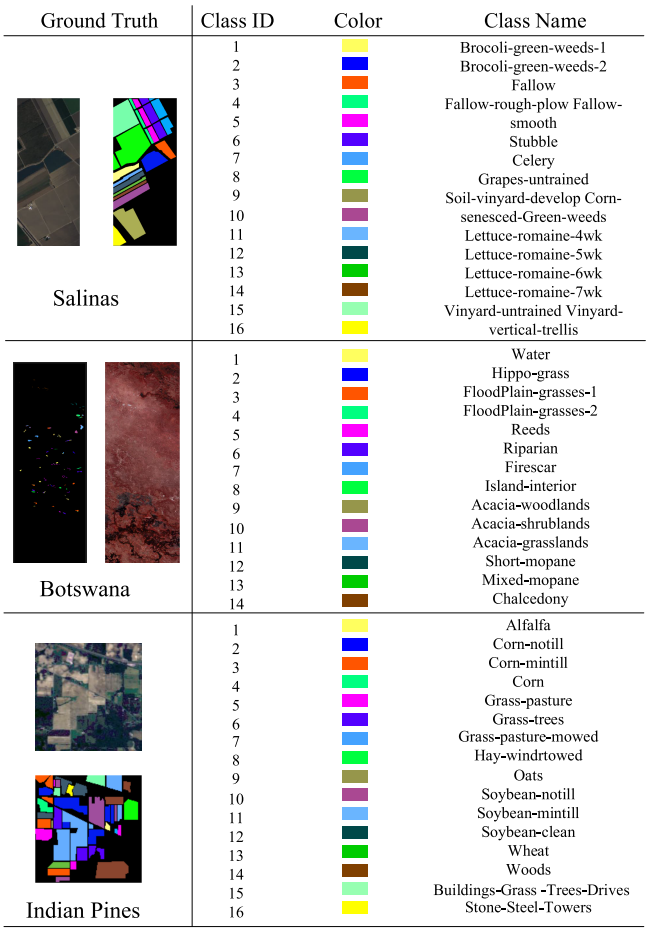


Fig. 4. Detailed information of three datasets for experiments.

The pseudo-color and its corresponding ground truth map are depicted in Fig. 4. We selected 29, 19, and 9 as its RGB channels.

The second dataset (Botswana), sourced from the EO-1 satellite of NASA, was collected between 2001 and 2004 over the Botswana region. The image has 242 bands with a pixel resolution of 30 m. These bands span the range of 400–2500 nm at 10 nm intervals. After eliminating noise bands, 145 candidate features were retained, including [10–55, 82–97, 102–119, 134–164, 187–220]. The dataset comprises 14 classes. The pseudo-color and the corresponding ground truth maps are presented in Fig. 4. We selected 35, 15, and 6 as its RGB channels.

The third dataset (Indian Pines), acquired by the AVIRIS sensor, contains the Indian Pines test site in Northwestern Indiana. After excluding water absorption bands, this image encompasses 200 spectral bands with 145 $\times$ 145 pixels. Its spectral coverage spans from 400–2500 nm, with a spatial resolution of 20 m. The pseudo-color and the corresponding ground truth map are shown in Fig. 4. We selected 32, 21, and 11 as its RGB channels.

The fourth dataset (Houston2013) was acquired from the University of Houston campus and the neighboring urban area. It consists of 144 spectral bands in the 380–1050 nm region with 1905 $\times$ 349 pixels. Its spatial resolution is 2.5 m.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE SALINAS DATASET (%)

Evaluation Metrics	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
NMI	73.56	78.91	79.86	66.20	64.84	82.93	74.12	62.22	66.80	72.94	**83.52**
ARI	54.66	63.43	59.22	45.53	43.21	66.57	55.41	44.22	49.37	53.85	**67.62**
AMI	72.62	72.29	71.57	66.17	64.81	82.91	74.10	62.19	66.78	72.92	**83.51**
FMI	59.28	67.39	68.06	54.13	49.14	72.30	60.17	50.65	58.89	59.07	**73.43**
OA	67.99	69.07	69.41	57.10	55.38	72.16	66.13	55.05	57.89	67.31	**75.28**

TABLE II
ACCURACIES IN EACH CLASS FOR DIFFERENT METHODS ON THE SALINAS DATASET (%)

Method	Class	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
PA	1	**98.51**	0	0	0	0	0	0	**100**	0	0	0
	2	83.90	98.42	96.22	87.98	97.88	**99.81**	96.94	99.03	92.86	85.83	98.03
	3	38.01	42.00	0.10	0	4.25	58.35	81.98	0.05	**85.17**	0	54.30
	4	**99.71**	0	99.35	0	83.36	99.43	93.69	85.94	90.32	99.50	89.89
	5	99.22	96.71	99.10	**99.89**	76.89	91.64	85.88	78.42	78.72	97.39	85.55
	6	99.44	99.39	99.39	80.90	**99.47**	94.06	97.47	95.45	49.89	96.89	97.30
	7	38.53	99.36	97.46	0.36	0.03	**99.47**	98.18	88.49	98.58	98.69	98.55
	8	58.72	97.18	**98.94**	82.97	52.86	94.53	35.52	51.45	91.70	58.58	97.06
	9	94.44	96.07	98.34	96.76	64.24	93.42	**98.86**	91.38	87.60	96.84	97.40
	10	56.35	59.46	63.79	45.15	24.67	**66.38**	54.97	0.34	0.52	0.03	63.00
	11	96.82	**98.31**	3.00	0	56.46	0.37	0	71.63	74.91	56.09	87.92
	12	0.16	45.36	48.26	71.67	67.20	99.38	2.02	0	0	99.07	**100**
	13	98.03	97.49	0	18.45	98.14	12.34	81.11	0.11	0	**98.80**	98.58
	14	79.53	0	92.90	63.08	89.07	**99.81**	90.84	0.28	0	86.17	85.98
	15	61.53	0	0	18.82	46.62	0.01	**79.88**	48.51	0.01	67.83	0
	16	0	62.48	67.63	72.72	66.52	74.16	4.65	3.87	40.56	0	**74.77**
UA	1	**99.85**	0	0	0	0	0	0	6.43	0	0	0
	2	56.98	64.65	64.02	52.55	38.26	**64.93**	64.81	63.90	63.12	61.43	64.58
	3	**97.41**	89.25	33.33	0	16.00	95.61	59.80	0.21	44.02	0	74.98
	4	98.65	0	**99.50**	0	72.90	94.74	91.65	23.53	97.30	95.13	93.30
	5	78.56	59.88	68.30	41.61	55.13	**99.55**	94.69	34.31	90.71	78.46	89.88
	6	99.52	99.95	99.87	100	99.75	99.81	100	99.92	100	**100**	99.79
	7	48.10	86.48	**97.98**	0.73	1.09	88.62	88.60	68.51	54.32	81.57	89.84
	8	73.09	58.52	54.78	51.27	**73.89**	59.77	64.52	62.81	53.13	70.33	59.51
	9	97.50	87.25	74.96	96.11	70.69	99.16	86.86	64.57	63.66	70.40	**99.18**
	10	61.69	78.05	89.59	**95.79**	20.11	58.68	52.92	47.83	32.08	33.33	93.48
	11	33.70	50.90	3.62	0	24.17	20.00	0	46.48	35.57	50.72	**99.05**
	12	2.73	**84.61**	83.11	45.94	36.94	70.79	6.51	0	0	57.67	81.72
	13	35.48	48.48	0	**86.22**	82.63	13.93	64.33	1.30	0	41.90	61.30
	14	55.84	0	66.49	**94.27**	84.64	36.06	82.03	1.51	0	52.78	54.96
	15	49.94	0	0	36.03	46.32	0.59	49.40	44.93	1.10	**51.30**	0
	16	0	51.98	86.30	99.02	85.49	92.86	4.58	54.69	31.84	0	**97.90**
OA		67.99	69.07	69.41	57.10	55.38	72.16	66.13	55.05	57.89	67.31	**75.28**

TABLE III
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE BOTSWANA DATASET (%)

Evaluation Metrics	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
NMI	62.44	67.14	60.66	67.22	54.94	75.38	56.35	45.58	62.07	63.95	**77.20**
ARI	42.45	49.81	36.45	49.81	51.68	59.61	33.06	24.34	39.91	45.76	**62.67**
AMI	62.04	66.87	59.62	66.87	66.78	63.59	55.89	44.99	61.74	63.59	**76.97**
FMI	48.07	54.60	44.74	54.60	41.19	63.42	39.07	31.73	46.16	50.78	**65.80**
OA	54.80	63.76	46.58	64.44	52.52	67.67	46.83	41.26	55.85	60.16	**73.89**

16 land-covers were labeled as ground truth. We selected 60, 28, and 13 as its RGB channels.

C. Clustering Performances Comparison

In this article, the clustering methods are categorized into centroid-based clustering methods (k-means), subspace-based clustering methods (USPEC [54], SGCNR [55], S^3C [56], S^5C [57], and S^2DMSC [58]), and deep learning-based clustering methods (CAE [59], VAE [60], DCN [26], and DEC [27]). Across all these methods, the number of clusters is set by referencing the ground truth in each dataset. Furthermore, for all the clustering techniques, the parameters are manually tuned to achieve their optimal values.

To evaluate the clustering performance, we utilize a serial of clustering metrics, including producer's accuracy (PA), user's accuracy (UA), overall accuracy (OA), fowlkes mallows index (FMI), adjusted rand index (ARI), adjusted mutual information (AMI), and normalized mutual information (NMI).

1) Quantitative Evaluation: From Tables I–VI, most of the subspace-based clustering methods significantly outperform prototype-based clustering method (k-means) on all the first

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ACCURACIES IN EACH CLASS FOR DIFFERENT METHODS ON THE BOTSWANA DATASET (%)

Method	Class	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
PA	1	68.52	100	33.70	91.48	52.59	100	88.89	92.22	100	98.52	**100**
	2	48.51	57.43	83.17	67.33	80.2	0	**83.17**	30.69	0	16.83	66.34
	3	0	96.02	0	79.28	82.47	0	39.44	55.78	75.30	**97.21**	86.06
	4	70.70	22.33	26.51	**88.84**	19.53	57.67	65.12	0	0	0	84.65
	5	80.67	**83.27**	73.23	66.91	81.04	56.88	0.74	32.34	0	80.67	61.71
	6	0	0	0	2.60	46.10	49.07	11.52	58.36	**60.22**	21.19	4.83
	7	94.21	96.91	97.30	81.85	76.45	100	50.97	44.02	53.28	87.26	**97.30**
	8	52.71	45.32	68.47	42.36	43.84	0.49	0	66.01	11.33	6.90	**99.51**
	9	55.10	73.89	70.70	76.43	52.55	**83.12**	28.98	21.97	62.42	63.69	72.93
	10	51.21	18.15	0	38.71	20.97	**97.58**	63.31	43.15	62.90	58.47	43.15
	11	**99.02**	88.20	95.74	75.41	63.93	86.56	34.10	38.69	68.52	55.41	64.92
	12	72.38	76.24	98.34	0	53.59	100	93.37	27.62	93.92	83.98	**100**
	13	21.64	57.84	0	95.90	35.82	88.81	72.01	0	76.87	56.72	**88.81**
	14	36.84	50.53	1.05	84.21	0	76.84	83.16	**88.42**	100	100	83.16
UA	1	100	**100**	100	98.41	100	100	100	100	99.63	98.52	96.43
	2	**92.45**	82.86	91.3	70.83	78.64	0	47.46	67.39	0	50.00	77.01
	3	0	56.97	0	**92.13**	56.71	0	33.00	22.54	49.35	53.74	56.84
	4	75.25	28.92	36.08	**90.95**	22.11	44.93	27.03	0	0	0	65.47
	5	41.49	46.09	53.10	71.15	65.47	58.40	3.39	55.41	0	**75.61**	64.59
	6	0	0	0	10.77	45.59	44.75	34.83	**58.15**	38.57	31.15	20.31
	7	96.44	99.60	91.30	99.53	60.74	70.77	90.41	66.28	99.28	99.56	**100**
	8	71.33	**92**	50.36	62.32	28.99	3.03	0	30.11	56.10	70	84.87
	9	73.00	**69.67**	47.44	55.68	65.74	67.62	27.91	19.17	35.83	42.11	48.11
	10	50.00	73.77	0	57.14	31.71	**78.32**	70.40	24.37	59.32	47.39	60.11
	11	50.25	65.93	36.45	51.45	51.32	**96.35**	45.22	77.12	65.52	50.00	93.40
	12	44.56	50.36	33.40	0	47.32	49.32	37.89	26.60	60.71	64.96	**95.77**
	13	23.58	69.51	0	46.22	49.74	71.9	53.46	0	47.58	56.93	**100**
	14	22.29	41.03	**100**	45.98	0	97.33	94.05	69.42	62.50	62.09	66.39
OA		54.80	63.76	46.58	64.44	52.52	67.67	46.83	41.26	55.85	60.16	**73.89**

TABLE V
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE INDIAN PINES DATASET (%)

Evaluation Metrics	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
NMI	43.86	44.70	46.36	39.03	43.05	50.31	38.80	39.09	32.39	40.24	**51.73**
ARI	21.95	22.48	26.45	20.24	23.54	**35.57**	21.25	25.15	14.95	22.49	34.07
AMI	42.17	42.87	45.83	38.75	42.78	50.06	38.51	38.80	32.09	40.01	**51.49**
FMI	30.71	31.28	36.64	28.83	32.40	44.36	30.65	34.66	24.61	37.07	**44.40**
OA	38.11	37.78	42.40	37.02	39.81	54.12	35.04	39.50	29.84	39.12	**55.51**

three datasets. k-means relies on the raw data features for clustering. In contrast, the fundamental premise of subspace-based clustering methods is based on the assumption that each category in the dataset corresponds to one low-dimensional subspace. These methods aim to achieve more accurate clustering by learning or estimating a subspace representation of data in which the data samples have better discriminative performance. From these tables, the performance of S^5C falls behind k-means. One potential explanation might be that S^5C with sparse representation and selective sampling fails to adequately capture the full nuances of the HSI data. In the deep learning-based methods, compared with CAE, DCN, and VAE, the separable clustering method (DEC) achieves better clustering accuracy. This is due to the iterative refinement approach adopted by DEC, which jointly optimizes the deep embedding representation and clustering tasks. Our proposed method, fully considering the related information of HSI data and utilizing the advanced transformer structure to extract high-level semantic information from different levels, has shown significant advancements. Furthermore, to enhance the robustness of the model against noise, we employ a dual-branch in both the shared autoformer module and the Siamese DFGM. This dual-branch approach is designed to separately process the original data and noisy data (with Gaussian noise) to improve the robustness of the network. The experimental results show that the proposed SDST not only takes into account the global correlation information between features but also fully utilizes the spectral information of HSI data, thus our method has obvious advantages in different scenarios. In particular, on the SA dataset, SDST exceeds 3.12% and 7.97% than the best comparison methods (i.e., S^2DMSC and DEC) in terms of OA.

2) Qualitative Evaluation: Figs. 5–7 present the visual classification maps of different methods on the first three datasets. From the figures, it can be observed that, compared with all the other benchmarked methods, the regions acquired by our proposed SDST are more uniform with the fewest misclassified pixels. This is particularly evident in the areas where the intermixed ground objects have similar spectral characteristics.

Taking the Salinas dataset as an example (see Fig. 5), the pixels for the land cover types of "Brocoli-green-weeds-1" and "Brocoli-green-weeds-2" are prone to be misclassified by most methods due to their overly similar spectral features. Most notably, our method effectively differentiates the two objects. Our method adopts an advanced hierarchical feature extraction method that can effectively identify and exploit relevant features in HSI data. This is especially evident when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ACCURACIES IN EACH CLASS FOR DIFFERENT METHODS ON THE INDIAN PINES DATASET (%)

Method	Class	k-means	USPEC	SGCNR	S^3C	S^5C	S^2DMSC	CAE	VAE	DCN	DEC	OUR
PA	1	11.11	0	18.52	1.85	11.11	0	0	0	**74.07**	0	0
	2	26.22	29.43	30.96	21.97	32.78	**48.40**	31.59	28.38	25.52	6.28	38.24
	3	35.37	**40.89**	3.12	34.65	32.61	8.75	33.69	9.71	8.27	0	26.99
	4	**41.45**	19.23	20.09	32.48	8.97	11.54	15.81	22.65	9.40	18.38	0
	5	48.69	39.24	44.87	60.36	**62.78**	31.59	14.89	51.91	48.09	45.47	52.17
	6	28.92	30.39	45.25	55.15	59.84	83.53	32.26	47.12	33.20	61.58	**87.26**
	7	0	0	0	50	76.92	0	69.23	11.54	0	0	**100**
	8	98.36	93.25	60.53	53.99	56.03	**98.98**	71.17	41.51	24.54	66.67	78.66
	9	0	55.00	70.00	0	20.00	**100**	0	0	0	0	0
	10	38.84	31.92	33.26	27.07	28.51	**50.62**	12.71	45.76	10.64	27.07	39.51
	11	38.01	40.44	67.46	36.02	41.69	67.99	36.75	64.59	37.36	74.84	**80.20**
	12	**29.97**	18.73	13.19	11.40	18.24	1.79	10.91	18.40	18.89	11.73	10.00
	13	90.57	95.75	95.28	77.83	83.49	93.87	32.55	26.42	53.30	0.47	**96.59**
	14	37.64	33.38	42.66	47.76	46.14	**83.46**	72.18	29.29	50.31	48.38	62.93
	15	15.79	20.79	23.95	26.05	8.16	14.74	21.05	17.89	21.32	26.84	**36.53**
	16	0	87.37	88.42	67.37	83.16	27.37	0	**89.47**	3.16	0	87.10
UA	1	5.94	0	14.29	0.12	3.61	0	0	0	**31.50**	0	0
	2	45.63	56.27	**56.63**	37.37	49.84	43.84	22.58	43.02	50.14	28.39	56.52
	3	18.19	18.58	53.70	21.86	15.64	20.00	20.06	35.37	12.21	0	**57.29**
	4	12.37	11.60	5.77	10.26	9.21	**27.27**	12.59	11.02	3.91	21.29	0
	5	34.47	30.09	31.81	34.68	30.32	**95.73**	22.09	29.97	38.74	30.92	45.32
	6	69.01	82.25	81.06	57.54	76.28	**91.36**	54.28	46.44	72.09	40.64	80.33
	7	0	0	0	8.02	7.38	0	4.84	0.82	0	0	**8.75**
	8	62.39	**88.72**	87.57	54.21	82.53	66.85	73.73	40.60	17.62	41.74	87.85
	9	0	2.86	8.00	0	2.19	**6.60**	0	0	0	0	0
	10	**56.54**	35.44	22.56	34.25	34.72	28.49	15.49	29.97	10.72	26.76	32.96
	11	47.33	48.90	49.09	52.02	48.47	**61.56**	48.58	55.21	37.71	39.92	52.87
	12	25.92	25.61	27.27	22.29	24.14	15.71	**40.85**	26.04	25.66	35.82	25.91
	13	32.76	33.5	40.81	70.82	49.30	54.22	18.60	26.17	12.14	**100**	66.00
	14	85.59	85.21	88.32	84.89	84.08	**88.74**	80.73	59.87	77.78	77.48	83.00
	15	20.07	34.05	17.17	23.63	19.62	30.27	28.67	19.05	22.50	25.12	**49.65**
	16	0	91.21	82.35	37.87	28.21	47.27	0	**97.70**	0.67	0	94.17
OA		38.11	37.78	42.40	37.02	39.81	54.21	35.04	39.50	29.84	39.12	**55.51**

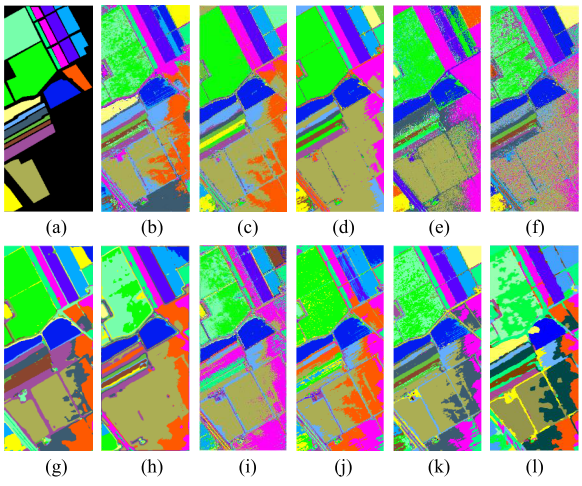


Fig. 5. Clustering maps of Salinas image. (a) Ground truth. (b) k-means. (c) USPEC. (d) SGCNR. (e) S^3C. (f) S^5C. (g) S^2DMSC. (h) CAE. (i) VAE. (j) DCN. (k) DEC. (l) OUR.

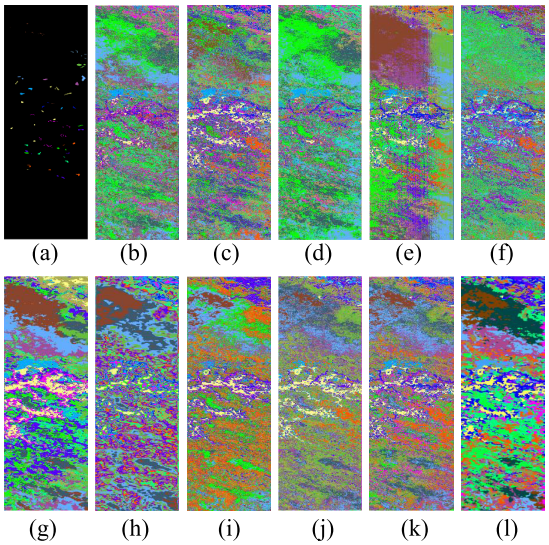


Fig. 6. Clustering maps of Botswana image. (a) Ground truth. (b) k-means. (c) USPEC. (d) SGCNR. (e) S^3C. (f) S^5C. (g) S^2DMSC. (h) CAE. (i) VAE. (j) DCN. (k) DEC. (l) OUR.

processing the Botswana dataset, which enables a clearer distinction between water and nonwater areas. By comparing clustering maps, we demonstrate the difference between our method and traditional methods in identifying feature edges on the Indian Pines dataset. The clustering maps clearly show that our method has significant advantages in edge sharpness and continuity on the first three datasets. The classification maps demonstrate that the proposed SDST can significantly enhance the discrimination ability for similar ground objects.

D. Parameter Analysis

SDST investigates several crucial parameters that have notable impacts on the clustering performance: parameters α and β, the number of superpixels N, the number of neighbor nodes k, and the number of transformer blocks L in autoformer. In this study, we employ cross-validation techniques to meticulously assess the sensitivity of each parameter. For

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

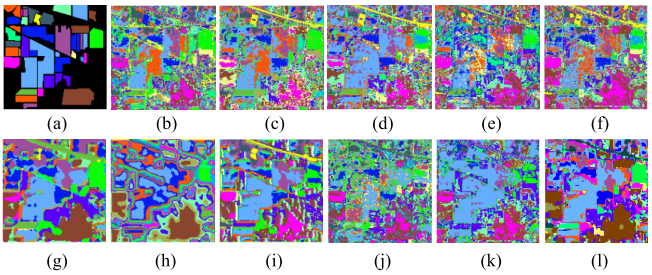


Fig. 7. Clustering maps of Indian image. (a) Ground truth. (b) k-means. (c) USPEC. (d) SGCNR. (e) S^3C. (f) S^5C. (g) S^2DMSC. (h) CAE. (i) VAE. (j) DCN. (k) DEC. (l) OUR.

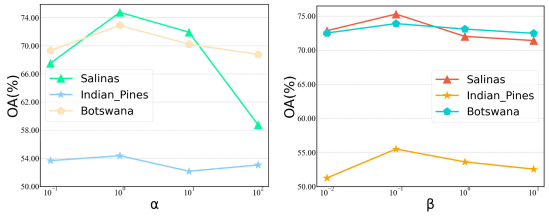


Fig. 8. Analysis of parameters α and β in the overall loss function.

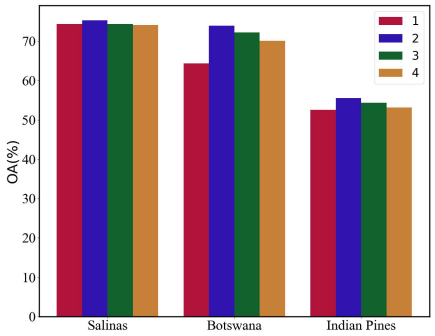


Fig. 9. OAs with different numbers of transformer block layers in autoformer.

each parameter, we provide corresponding variation curves of OA on the Indian Pines, Salinas, and Botswana datasets.

1) Parameters α and β: In the proposed method, the overall loss function primarily has two parameters: α and β. The OAs are illustrated in Fig. 8. In our experiment, we initially set β as 0 to analyze the optimal value of α. As shown in Fig. 8, the peak OA is achieved when α is designated as 1. Subsequently, with α being fixed to 1, we analyze the most suitable value for β. From Fig. 8, the best result is obtained when β is set to 0.1. Therefore, by setting α and β as 1 and 0.1, our method can achieve satisfactory performance.

2) Number of Transformer Blocks L: In SDST, the autoformer is designed to capture multilevel features of hyperspectral data. The transformer module is employed to extract global features from different levels. The number of layers in the transformer has a significant impact on the clustering performance. On the first three datasets, the optimal clustering performance is achieved when two transformer layers are used in Fig. 9.

3) Number of Superpixels N: The parameter sensitivity results for N are illustrated in Fig. 10. On the Indian Pines dataset, N varies from 1000 to 3000 with an interval of 500. On the Salinas dataset, N ranges from 100 to 2000 with an interval of 500. On the Botswana dataset, N shifts from 3000 to 7000 with a step of 1000. The OAs increase on all the first three datasets with the augmentation of N. However, beyond a certain threshold of N, there is not a discernible improvement in clustering accuracy. A larger N signifies smaller superpixels, hence, we can retain more locally detailed features of HSI for clustering. Nonetheless, a larger N results in an increased number of graph nodes, and an excessive number of these nodes poses challenges for network feature extraction and increases computational complexity. Therefore, the optimal values of N for Salinas, Botswana, and Indian Pines are determined to be 1000, 6000, and 2000, respectively.

TABLE VII
SENSITIVITY ANALYSIS OF THE PARAMETER k ON SDST

Datastes	OA(%)/k	OA(%)/k	OA(%)/k	OA(%)/k
Salinas	72.22/3	74.86/5	75.28/7	74.16/9
Botswana	69.86/11	69.12/14	73.89/17	71.31/20
Indian Pines	52.64/8	53.21/10	55.51/12	54.89/14

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART CLUSTERING METHODS ON THE HOUSTON2013 DATASET

DAtastes	k-means	ESC	IDEC	TMPCC	OUR
OA	39.90	37.69	44.14	47.26	**49.06**
Kappa	35.04	32.77	39.17	43.00	**45.10**
NMI	44.51	36.43	47.82	**51.09**	50.94
ARI	23.78	21.08	26.28	**31.94**	28.64
Purity	43.02	39.52	46.48	49.46	**49.60**

4) Number of Neighbor Nodes k: To analyze the number of neighbor nodes, we search for the optimal number of neighbor nodes and display the corresponding results in Table VII. It is worth noting that different hyperspectral datasets exhibit varying levels of complexity and possess different feature spaces. The Botswana and Indian Pines datasets are characterized by intricate structures and diverse spectral signatures, whereas the Salinas dataset exhibits a relatively simpler spatial structure. Consequently, the clustering algorithm requires a larger number of neighbor nodes to accurately capture the local structure of the Botswana and Indian Pines datasets, while a smaller number is used for the Salinas dataset. Remarkably, our results indicate that the best clustering results are achieved when setting k to 7, 12, and 17 for the Salinas, Botswana, and Indian Pines datasets, respectively.

5) Experiments on Large-Scale Dataset: As shown in Table VIII, to verify the clustering performance of the designed method on a large-scale HSI dataset, we used Houston2013 to test SDST. To evaluate clustering performance, we utilize a series of clustering metrics, including OA, Kappa, NMI, and ARI. By comparing with k-means [7], ESC [61], IDEC [62], and TMPCC [63] methods, it can be seen that SDST also shows excellent clustering performance on the large-scale HSI dataset.

E. Ablation Study

In this experiment, the impact of autoformer and DFGM for HSI clustering is assessed on the Salinas, Botswana, and Indian Pines datasets. The results are presented in Table IX. These methods are simplified from SDST-1 to SDST-7. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

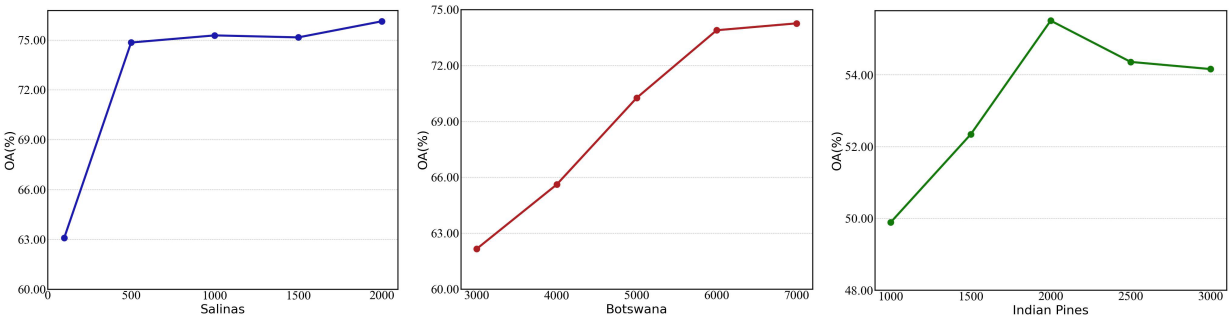


Fig. 10. Different N values on the Salinas, Botswana, and Indian Pines datasets.

TABLE IX
ABLATION STUDY RESULTS ON THE SALINAS, BOTSWANA, AND INDIAN PINES DATASETS (%)

DAtastes	SDST-1	SDST-2	SDST-3	SDST-4	SDST-5	SDST-6	SDST-7
Autoformer	-	-	-	✓	✓	-	-
Shared Autoformer	-	✓	-	-	-	✓	✓
Single-Former Graph Module	-	-	-	✓	-	✓	-
Dual-Former Graph Module	-	-	✓	-	✓	-	✓
Salinas	55.90	72.35	71.59	73.39	74.20	74.61	75.28
Botswana	56.79	66.93	69.03	70.04	71.49	70.59	73.89
Indian Pines	42.03	52.63	51.26	50.01	53.02	53.56	55.51

results reveal that SDST performs superiorly. Specifically, the clustering accuracy of SDST-2 is higher than that of SDST-1, indicating that the shared Autoformer effectively enhances clustering accuracy. Compared with SDST-1, SDST-3 yields better results, confirming the significant role of DFGM in the proposed method. A comparison among SDST-4, SDST-5, and SDST-6 shows that the shared Autoformer and the Siamese DFGM are beneficial for model clustering. Ultimately, the OA values of SDST-7 surpass those from SDST-1 to SDST-6, demonstrating the superiority of SDST. Hence, autoformer and DFGM are instrumental in improving the clustering accuracy of SDST.

IV. CONCLUSION

In this article, we introduce a novel SDST method to boost the clustering performance for HSI datasets. We design an autoencoder to capture multilevel features and use the transformer to obtain global attribute information. In addition, we innovatively designed the Siamese DFGM to learn long-dependency graph features, which enhances the local details and global dependencies of features. Finally, we integrate the shared autoformer module and the Siamese DFGM with a joint optimization module for network training. Experimental results demonstrate the effectiveness of the method on four widely used datasets. For SDST, the network structure is very complex. In the future, we will develop a lightweight network based on SDST.

REFERENCES

[1] W. Sun et al., "MLR-DBPFN: A multi-scale low rank deep back projection fusion network for anti-noise hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522914.

[2] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS^{3}LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[3] G. Yang, K. Huang, W. Sun, X. Meng, D. Mao, and Y. Ge, "Enhanced mangrove vegetation index based on hyperspectral images for mapping mangrove," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 236–254, Jul. 2022.

[4] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3250450.

[5] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[6] R. Rada, E. Bedalli, S. Shurdhi, and B. Çiço, "A comparative analysis on prototype-based clustering methods," in *Proc. 12th Medit. Conf. Embedded Comput. (MECO)*, Jun. 2023, pp. 1–5.

[7] A. M. Ikotun, A. E. Ezugwu, L. Abualigah, B. Abuhaija, and J. Heming, "K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data," *Inf. Sci.*, vol. 622, pp. 178–210, Apr. 2023.

[8] H. Jia, S. Ding, X. Xu, and R. Nie, "The latest research progress on spectral clustering," *Neural Comput. Appl.*, vol. 24, nos. 7–8, pp. 1477–1486, Jun. 2014, doi: 10.1007/s00521-013-1439-2.

[9] R. Vidal, "Subspace clustering," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 52–68, Mar. 2011.

[10] S. Huang, H. Zhang, and A. Pižurica, "Hybrid-hypergraph regularized multiview subspace clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3074184.

[11] R. Vidal, Y. Ma, and S. Sastry, "Generalized principal component analysis (GPCA)," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 12, pp. 1945–1959, Dec. 2005.

[12] J. Yin and S. Sun, "Incomplete multi-view clustering with reconstructed views," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 3, pp. 2671–2682, Mar. 2023.

[13] G. Liu, Z. Lin, and Y. Yu, "Robust subspace segmentation by low-rank representation," in *Proc. 27th Int. Conf. Int. Conf. Mach. Learn.*, 2010, pp. 663–670.

[14] Q. Zhang and Z. Miao, "Subspace clustering via sparse graph regularization," in *Proc. 4th IAPR Asian Conf. Pattern Recognit. (ACPR)*, Nov. 2017, pp. 214–219.

[15] Z. Kang, Z. Lin, X. Zhu, and W. Xu, "Structured graph learning for scalable subspace clustering: From single view to multiview," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 8976–8986, Sep. 2022.

[16] T. Wang, H. Cai, X. Zhang, L. Lan, X. Huang, and Z. Luo, "Graph-Laplacian correlated low-rank representation for subspace clustering," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3748–3752.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

[17] M. Yin, S. Xie, Z. Wu, Y. Zhang, and J. Gao, "Subspace clustering via learning an adaptive low-rank graph," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 3716–3728, Aug. 2018.

[18] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3127536.

[19] S. Liu and H. Wang, "Graph convolutional optimal transport for hyperspectral image spectral clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3203481.

[20] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[21] N. Huang, L. Xiao, Y. Xu, and J. Chanussot, "A bipartite graph partition-based coclustering approach with graph nonnegative matrix factorization for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3097358.

[22] N. Huang, L. Xiao, J. Liu, and J. Chanussot, "Graph convolutional sparse subspace coclustering with nonnegative orthogonal factorization for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512016.

[23] Y. Ma, C. Li, X. Mei, C. Liu, and J. Ma, "Robust sparse hyperspectral unmixing with $\ell_{2,1}$ norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1227–1239, 2016.

[24] H. Zhai, H. Zhang, P. Li, and L. Zhang, "Hyperspectral image clustering: Current achievements and future lines," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 35–67, Dec. 2021.

[25] E. Min, X. Guo, Q. Liu, G. Zhang, J. Cui, and J. Long, "A survey of clustering with deep learning: From the perspective of network architecture," *IEEE Access*, vol. 6, pp. 39501–39514, 2018.

[26] B. Yang, X. Fu, N. D. Sidiropoulos, and M. Hong, "Towards k-means-friendly spaces: Simultaneous deep learning and clustering," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3861–3870.

[27] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 478–487.

[28] R. Lopez, P. Boyeau, N. Yosef, M. Jordan, and J. Regier, "Decision-making with auto-encoding variational Bayes," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 5081–5092.

[29] Y. Zhao and X. Li, "Deep spectral clustering with regularized linear embedding for hyperspectral image clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509311.

[30] Z. Mei and Z. Yin, "Self-supervised learning of discriminative spatial–spectral features for hyperspectral images clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3168722.

[31] K. R. Shahi, P. Ghamisi, B. Rasti, P. Scheunders, and R. Gloaguen, "Unsupervised data fusion with deeper perspective: A novel multisensor deep clustering algorithm," *IEEE J. Sel. Topics Appl. Earth Obser. Remote Sens.*, vol. 15, pp. 284–296, 2022.

[32] J. Sun et al., "Deep clustering with intraclass distance constraint for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4135–4149, May 2021.

[33] F. Wu, A. H. Souza, T. Zhang, C. Fifty, T. Yu, and K. Q. Weinberger, "Simplifying graph convolutional networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6861–6871.

[34] D. Bo, X. Wang, C. Shi, M. Zhu, E. Lu, and P. Cui, "Structural deep clustering network," in *Proc. Web Conf.*, Apr. 2020, pp. 1400–1410.

[35] Y. Zeng, X. Zhou, J. Rao, Y. Lu, and Y. Yang, "Accurately clustering single-cell RNA-seq data by capturing structural relations between cells through graph convolutional network," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2020, pp. 519–522.

[36] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[37] B. Wu et al., "Visual transformers: Where do transformers really belong in vision models?" in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 579–589.

[38] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[39] W. Wang et al., "Crossformer: A versatile vision transformer hinging on cross-scale attention," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Oct. 2021, pp. 1–16.

[40] X.-B. Nguyen, D. T. Bui, C. N. Duong, T. D. Bui, and K. Luu, "Clusformer: A transformer based clustering approach to unsupervised large-scale face and visual landmark recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10842–10851.

[41] Y. Wang et al., "Self-supervised video-centralised transformer for video face clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 1–16, Feb. 2023.

[42] Z. Xue, X. Yu, A. Yu, B. Liu, P. Zhang, and S. Wu, "Self-supervised feature learning for multimodal remote sensing image land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3190466.

[43] Z.-Y. Li, S. Gao, and M.-M. Cheng, "SERE: Exploring feature self-relation for self-supervised transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15619–15631, Dec. 2023.

[44] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3310489.

[45] S. Huang, H. Zhang, and A. Pižurica, "A structural subspace clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3102422.

[46] S. Subudhi, R. N. Patro, P. K. Biswal, and F. Dell'Acqua, "A survey on superpixel segmentation as a preprocessing step in hyperspectral image analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5015–5035, Apr. 2021.

[47] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[48] Y. Ding et al., "AF2GNN: Graph convolution with adaptive filters and aggregator fusion for hyperspectral image classification," *Inf. Sci.*, vol. 602, pp. 201–219, Jul. 2022.

[49] S. Huang, H. Zhang, J. Xue, and A. Pižurica, "Heterogeneous regularization-based tensor subspace clustering for hyperspectral band selection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 1–15, Mar. 2022.

[50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[51] C. Ying et al., "Do transformers really perform badly for graph representation?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 28877–28888.

[52] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2008.

[53] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 1–27, 2008.

[54] D. Huang, C.-D. Wang, J.-S. Wu, J.-H. Lai, and C.-K. Kwoh, "Ultra-scalable spectral clustering and ensemble clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 6, pp. 1212–1226, Jun. 2020.

[55] R. Wang, F. Nie, Z. Wang, F. He, and X. Li, "Scalable graph-based clustering with nonnegative relaxation for large hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7352–7364, Oct. 2019.

[56] X. Peng, L. Zhang, and Z. Yi, "Scalable sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 430–437.

[57] S. Matsushima and M. Brbic, "Selective sampling-based scalable sparse subspace clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–10.

[58] N. Huang, L. Xiao, Q. Liu, and J. Chanussot, "S^2DMSC: A self-supervised deep multilevel subspace clustering approach for large hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512817.

[59] F. Li, H. Qiao, and B. Zhang, "Discriminatively boosted image clustering with fully convolutional auto-encoders," *Pattern Recognit.*, vol. 83, pp. 161–173, Nov. 2018.

[60] K.-L. Lim, X. Jiang, and C. Yi, "Deep clustering with variational autoencoder," *IEEE Signal Process. Lett.*, vol. 27, pp. 231–235, 2020.

[61] C. You, C. Li, D. P. Robinson, and R. Vidal, "Self-representation based unsupervised exemplar selection in a union of subspaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2698–2711, May 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.

[62] X. Guo, L. Gao, X. Liu, and J. Yin, "Improved deep embedded clustering with local structure preservation," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Aug. 2017, pp. 1753–1759.

[63] Y. Cai, Z. Zhang, P. Ghamisi, B. Rasti, X. Liu, and Z. Cai, "Transformer-based contrastive prototypical clustering for multimodal remote sensing data," *Inf. Sci.*, vol. 649, Nov. 2023, Art. no. 119655.

Fulin Luo (Senior Member, IEEE) received the B.E. degree in mechanical engineering and automation from Southwest Petroleum University, Chengdu, China, in 2011, and the M.E. and Ph.D. degrees in instrument science and technology from Chongqing University, Chongqing, China, in 2013 and 2016, respectively.

He was an Associate Researcher and a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, from 2017 to 2021. He was a Research Fellow with Nanyang Technological University, Singapore, from 2020 to 2021. He has been a Professor with the College of Computer Science, Chongqing University, since 2022. His research interests are remote sensing processing, computer vision, and biomedical analysis.

Yi Liu received the B.E. degree in computer science and technology from Northeast Forestry University, Harbin, Heilongjiang, China, in 2022. He is currently pursuing the M.E. degree in electronic information with Chongqing University, Chongqing, China.

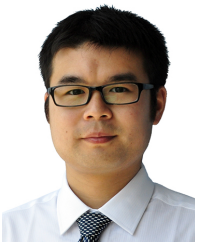
His research interests include hyperspectral clustering and self-supervised learning.

Yule Duan received the B.E. degree in measurement and control technology and instruments from Tianjin Polytechnic University, Tianjin, China, in 2016, and the Ph.D. degree in instrument science and technology from Chongqing University, Chongqing, China, in 2021.

He is currently a Lecturer with the College of Information Science and Engineering, Henan University of Technology, Zhenzhou, China. His research interests are image processing, remote sensing, and computer vision.

Tan Guo (Member, IEEE) received the M.E. degree in signal and information processing and the Ph.D. degree in communication and information systems from Chongqing University, Chongqing, China, in 2014 and 2017, respectively.

He has been a Lecturer with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, since 2018. He was a Post-Doctoral Fellow with the Macau University of Science and Technology, Macau, China, from 2020 to 2022. His research interests include computer vision, pattern recognition, and machine learning.



Lefei Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively.

He was a Big Data Institute Visitor with the Department of Statistical Science, University College London, London, U.K., and a Hong Kong Scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China. He is currently a Professor with the School of Computer Science, Wuhan University, and also with the Hubei Luojia Laboratory, Wuhan. His research interests include pattern recognition, image processing, and remote sensing.

Bo Du (Senior Member, IEEE) received the B.S. and Ph.D. degrees in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, in 2005 and 2010, respectively.

He is currently a Professor with the School of Computer Science, Institute of Artificial Intelligence, Wuhan University, where he is also the Director of the National Engineering Research Center for Multimedia Software. His major research interests include pattern recognition, hyperspectral image processing, and signal processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:52:10 UTC from IEEE Xplore. Restrictions apply.